

GATEWAY FIRMWARE · FANSTEL BLG20X

From Bluetooth tag to your platform, over cellular or satellite.

Firmware that listens to your Bluetooth devices, keeps what matters in flash, and delivers it over LTE-M, NB-IoT or satellite NB-NTN, choosing the cheapest uplink that works.

- ✓ Satellite NB-NTN registered and acknowledged on the hardware
- ✓ Commissioned from a browser, with no cable and no app
- ✓ Signed updates that roll back unless they confirm



Fanstel BLG20x · the sealed enclosure and the board inside

2

processors: nRF9151 for cellular and satellite, nRF54LM20B for Bluetooth LE

3

uplinks in one ladder: LTE-M, NB-IoT and satellite NB-NTN, cheapest first

128^B

satellite datagram, carrying a status and two alerts

~1^{kB}

sent per LTE-M cycle of up to eight records, measured on the hardware

FOR CTOS

Prove it early. Ship in weeks, not quarters.

- Start from firmware already running on this board, instead of a blank project and a long, fail-late road.
- No platform to rent: your devices, your network, your data platform.
- Every figure in this brief was measured or verified on the hardware, and dated.

FOR ARCHITECTS

One record shape, whatever the uplink.

- The deployment class fixes transport and encoding, so a store written on LTE-M is sent unchanged over satellite.
- Confirmable CoAP: records leave the gateway only when your platform acknowledges them.
- Alarms are never overwritten, and every dropped record is counted.

FOR EMBEDDED DEVELOPERS

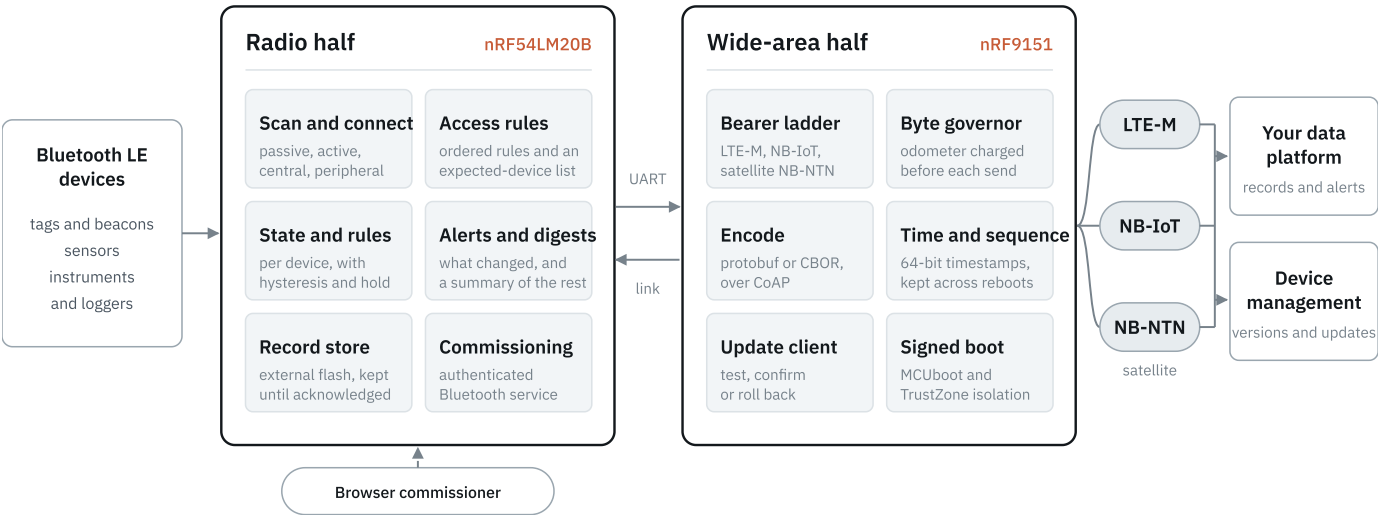
A board that is already known.

- nRF Connect SDK v3.4.0, Zephyr RTOS and MCUboot on both processors.
- Commission, self-test and diagnose from a browser over Bluetooth.
- Adsum IoT Coder holds the board knowledge, so you extend the firmware instead of rediscovering the hardware.

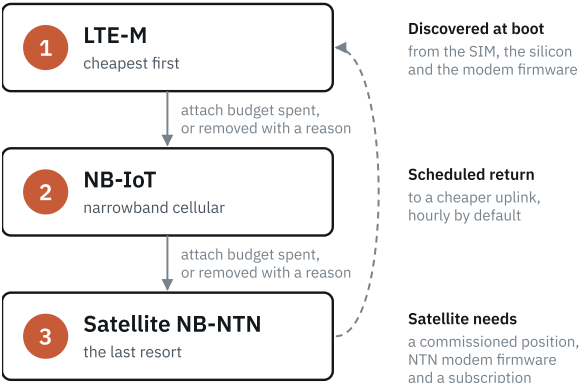
HOW IT WORKS

Two processors, one gateway.

The radio half listens and remembers; the wide-area half chooses the uplink and delivers. One serial link joins them and carries records, commands and the radio half’s updates.



The bearer ladder



Why an uplink is removed, and reported

- No access on the SIM
 - Modem firmware without satellite
 - Disallowed by configuration
 - Roaming refused
- Silicon unsupported
 - No satellite subscription
 - Locked network absent

Deployment class

CLASS	UPLINKS	TRANSPORT AND ENCODING
TN	LTE-M, NB-IoT	CoAP; protobuf or CBOR
NTN	Satellite only	CoAP; protobuf or CBOR
TN+NTN	The full ladder	One transport and encoding across the switch, so stored records are never re-encoded

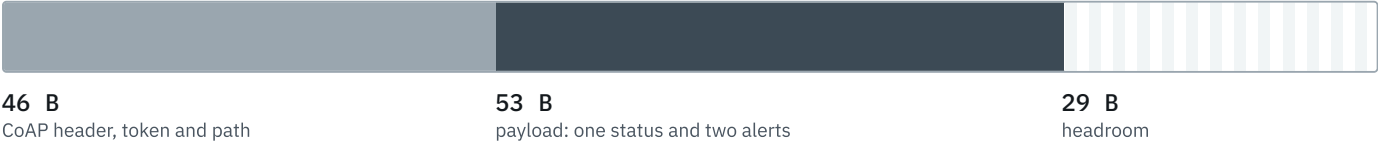
What reaches your platform

Digest	One record per device per window, on terrestrial uplinks	52 B
Alert	Raised with hysteresis and hold, sent with its age, at least once	13 B
Status	The heartbeat: devices seen and missing, store depth, uplink, quota	30 B
Health	Once a day on terrestrial uplinks: resets, attaches, drops	76 B
LTE-M cycle	Up to eight records in one confirmable POST	1,028–1,066 B

Sizes in protobuf, measured. CBOR is also supported.

Anatomy of a satellite datagram

128 B: the largest datagram the satellite uplink accepts

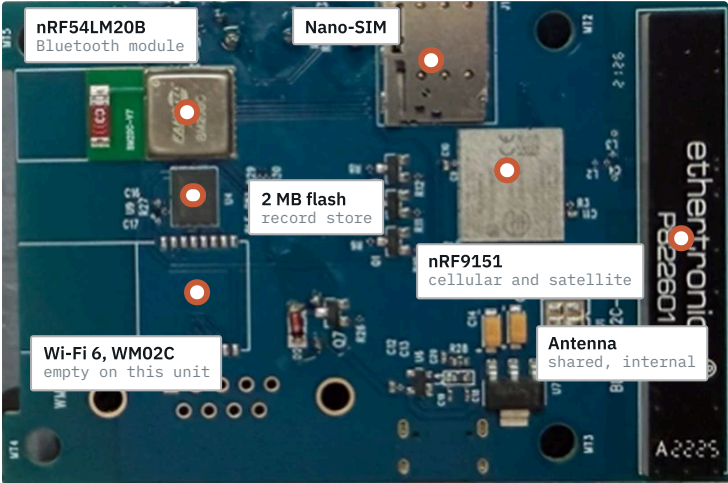


One real datagram, sent and acknowledged over a geostationary NB-NTN service on 13 September 2026.

FOR DEVELOPERS

Built on the SDK you already use.

Two Zephyr applications on nRF Connect SDK v3.4.0, one per processor, each behind its own MCUboot. Nothing exotic to learn, and nothing to rediscover about the board.



BLG20XE02C board, evaluation configuration

Flash used, read from the shipped images

nRF9151 application primary slot		393.3 / 448 KB
nRF9151 bootloader		29.6 / 64 KB
nRF54LM20B application slot 0		352.0 / 512 KB
nRF54LM20B bootloader		53.6 / 62 KB

Each bar is the partition; the fill is what the image uses.

Firmware at a glance

Processors	nRF9151 wide-area half; nRF54LM20B radio half
SDK	nRF Connect SDK v3.4.0, Zephyr RTOS, sysbuild, MCUboot on both halves
Between halves	UART framed with a sequence and CRC-16; a heartbeat every second; buffering continues if the link drops
Bluetooth LE	Passive and active scanning; central with three sessions and one kept for commissioning; a peripheral data-in service; LE Secure Connections only
Access	32 ordered rules ending in drop, count or quarantine; an expected-device list of 128; per-device rate ceilings
Uplink	LTE-M, NB-IoT, satellite NB-NTN; confirmable CoAP over UDP; protobuf or CBOR
Data	One record shape; 64-bit timestamps; a gateway sequence that survives reboots
Store	External flash, released only on acknowledgement; alarms never overwritten
Updates	Signed images; test boot; confirm behind a health gate; automatic roll-back; the radio half updated through the wide-area half
Commissioning	A web page over Web Bluetooth; every write validated on the unit and refused with a reason

Commissioning, in five steps

01 Connect Pair over LE Secure Connections from a phone's browser.	02 Position The phone's fix, typed, or surveyed, with its uncertainty.	03 Credentials Name the unit and give it its data platform token.	04 Commission Class, network policy, rules and expected devices.	05 Verify Read the unit back and compare it field by field.
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Recovery, built in

A processor hangs Its hardware watchdog reboots it and the cause is reported.
An update does not confirm The previous image is restored at the next boot.
Power is lost Acknowledged records are already in flash; the byte odometer is written before each send.

The link between the halves drops Both keep working; the store keeps filling; neither reboots.
A configuration fails three boots The previous configuration returns, and the roll-back is reported.
The modem is barred from attaching The unit reports it and waits; it never answers with a reset.

THE HARDWARE

Fanstel BLG20x, open hardware.

Fanstel designs, makes and sells the gateway, and publishes its design.
Adsum Networks writes, verifies, signs and licenses the firmware.

Cellular and satellite	nRF9151 SiP: LTE-M and NB-IoT (Release 14), NB-NTN (Release 17); Arm Cortex-M33; 1 MB flash, 256 KB RAM
Bluetooth LE	nRF54LM20B in a Fanstel BM20 module, Bluetooth 6.0 capable; Arm Cortex-M33; 2 MB non-volatile memory, 512 KB RAM; module options with or without a power amplifier
Storage	2 MB external flash, on the Bluetooth side
Antenna and SIM	One internal cellular antenna, shared by terrestrial and satellite; nano-SIM, with an eSIM footprint
Wi-Fi 6	Fanstel WM02C module (nRF7002): fitted on evaluation boards, empty on the unit shown here; this firmware version does not use it
Positioning	No GNSS on the BLG20XE02C, and no GNSS antenna: the position is set at commissioning
Controls	Reset and user buttons, two RGB indicators, a sounder; one SWD header per processor
Power	USB-C, the only power input; no battery
Enclosure	IP67; -40 to +80 °C (hardware maker's rating)



Sealed enclosure; indicators visible through the lid

Where to buy

The BLG20x is designed, made and sold by Fanstel. Hardware and pricing: fanstel.com

Proven on the hardware

10 Sep 2026	Commissioned from a phone's browser over Bluetooth
11 Sep 2026	Update applied, rolled back and confirmed behind the health gate
13 Sep 2026	Satellite NB-NTN registered, sent and acknowledged on a geostationary service
13 Sep 2026	Store held a satellite morning and drained in order, with no re-encoding
13 Sep 2026	LTE-M cycle delivered and acknowledged on the release image
13 Sep 2026	Access rules, rate ceilings and the expected-device list
13 Sep 2026	Configuration rolled back after three failed boots

In this version

- NB-IoT is in the ladder and not yet attached on this hardware.
- Downloading an update over the uplink is not yet proven end to end; evaluation units are updated over SWD.
- The modem firmware is installed before deployment and cannot be updated in the field.
- Wi-Fi is not used by this firmware version, and the hardware has no GNSS.
- Evaluation images: production signing keys are not yet in place, so no security claim is made for these images yet.
- Radio, capacity and power figures are published for the shipping configuration once it is confirmed.

Build it yourself

Adsum IoT Coder builds the full gateway from scratch much faster because it knows the hardware, and for licence holders the firmware too.

License ours

Start from the demo image pair in early access. The production image and the source, per module, are licensed on request.

Have it built

We build the variant you need, with a lead time and a price range on request.

Evaluate it on your own desk.

Ask for the demo image pair and the commissioner, flash both halves in minutes, and see your devices on your platform.

Ask for an evaluation image

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